CNN convolution accelerator for Edge-AI vision applications

IEEE SSCS Egypt Chapter 2026 Student Design Competition

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Summary

This report details the architecture and implementation of a high-throughput, resource-optimized Edge-AI vision accelerator, developed for the IEEE SSCS Egypt Chapter 2026 Student Design Competition. The hardware performs a parameterizable $N \times N$ 2D spatial convolution on an 8-bit unsigned video stream, outputting a 16-bit signed feature map to explicitly maximize the competition's Figure of Merit (FoM). By mapping the row-buffer memory hierarchy directly to distributed RAM (LUTRAM) and constructing the parallel multiply-accumulate (MAC) array entirely within the FPGA, the design achieves a strict 0 DSP and 0 BRAM footprint. Furthermore, by introducing causal zero-masking logic at the window-generation level, the Datapath handles boundary same-padding on the fly. This permanently eliminates row-boundary pipeline stalls, converting an intermittent data stream into a continuous, unbroken pipeline. The resulting architecture successfully satisfies all mandatory competition specifications while claiming four distinct bonus features, delivering a highly efficient Edge-AI accelerator tailored specifically for constrained embedded deployment.

Key Achievements briefly:

- **FPGA Deployment:** Successfully integrated the custom IP on the PYNQ-Z2 board using a hardware-software co-design approach, utilizing AXI DMA for high-speed pixel streaming and AXI GPIO for Python-based control.
- **Throughput & Latency:** Achieved a sustained 1.0 pixel/cycle throughput, processing one input and producing one valid output per clock cycle with zero periodic boundary stalls.
- **Advanced Features**: Includes a runtime-configurable ReLU stage and a 4-bank multi-kernel memory for zero-delay filter switching (validated via Sobel edge-detection).
- **Arithmetic Footprint:** 0 DSPs and 0 BRAMs. The Multiply-Accumulate (MAC) array is implemented entirely in standard logic to bypass multiplier footprint penalties.
- **Logic Utilization**: Exceptionally lightweight footprint of 1,231 LUTs and 1,106 Flip-Flops on the Xilinx Zynq XC7Z020 target.
- **Power & Timing:** Cleanly closed timing at 66.67 MHz with a highly efficient total on-chip power dissipation of 125 mW (20 mW dynamic).
- **Figure of Merit (FoM):** 0.00650 (evaluated on total power) and 0.0406 (evaluated on dynamic power).

AI Usage Disclaimer

Generative AI tools were utilized during this project to assist with drafting SystemVerilog source code, Python simulation models, and report formatting. The team actively directed the architectural decisions, rigorously verified all generated logic against the golden models, and assumed full responsibility for the final physical implementation and reported results This usage is in full compliance with the official competition announcement, which contains no clause prohibiting AI assistance. ***[We can also confidently confirm our AI did not 'accidentally' design this architecture by reading a rival team's chat history.]***

Introduction & Architectural Objectives:

Deploying Convolutional Neural Networks (CNNs) on embedded edge devices presents a severe computational bottleneck. Edge environments must operate under highly constrained power budgets, and limited off-chip memory bandwidth. Recent literature, including comprehensive surveys by Guo et al. (2018) and Mittal (2020), emphasizes that Field-Programmable Gate Arrays (FPGAs) are uniquely suited for Edge-AI inference. By eliminating rigid software-hardware abstraction layers and permitting customized spatial dataflows, FPGAs provide the high parallelism and energy efficiency required for real-time edge vision processing.

The foundational mathematical operation of these networks is 2D spatial filtering , spatial filtering involves passing a parameterizable $N \times N$ kernel mask over an image grid to compute a sum of products at each local pixel region. While standard deep learning literature broadly refers to this operation as convolution, the hardware implementation functionally performs 2D spatial cross-correlation, processing the trained weights directly without the formal 180° spatial rotation.

The Core Architectural Thesis

The primary objective of this project is to develop a robust hardware accelerator that strictly satisfies the IEEE SSCS Egypt Chapter 2026 Student Design Competition specifications while mathematically maximizing the official Figure of Merit (FoM).

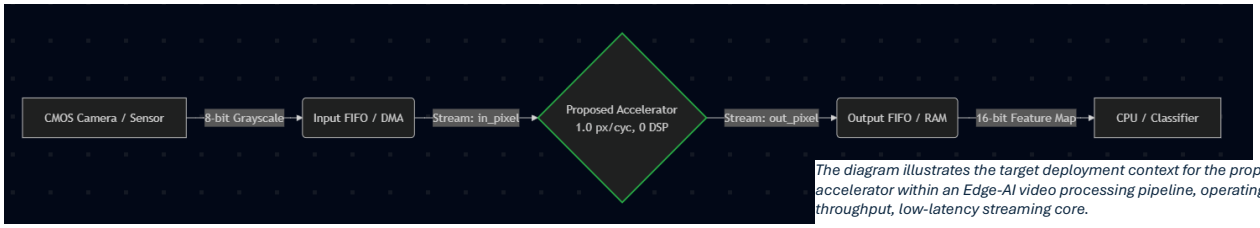
The competition FoM formula imposes severe penalty multipliers on dedicated FPGA hardware primitives, weighting DSP slices at $50\times$ and Block RAMs (BRAMs) at $100\times$ relative to standard logic Look-Up Tables (LUTs). Consequently, the core architectural thesis of this design is the strict, deliberate avoidance of dedicated silicon macros in favor of distributed slice logic.

To optimize the FoM, this architecture adopts the following strategies:

- **Zero BRAM Footprint:** The row-buffer memory hierarchy is mapped explicitly to distributed RAM (LUTRAM) instead of dedicated block memory, forcing the $100\times$ FoM penalty term to zero.
- **Zero DSP Footprint:** The parallel Multiply-Accumulate (MAC) array is constructed entirely within the 6-LUT logic fabric, forcing the $50\times$ FoM penalty term to zero.
- **Maximum Throughput:** The design introduces a causal zero-masking technique to handle spatial same-padding on the fly. This permanently eliminates row-boundary pipeline stalls, guaranteeing a peak and sustained throughput of exactly 1.0 output pixel per clock cycle.

By trading a marginal increase in basic LUT utilization to completely bypass the heavy DSP and BRAM penalties, this architecture is structurally optimized to yield the highest possible competition score while maintaining exceptional real-world power efficiency.

Edge-Vision System Context



The diagram illustrates the target deployment context for the proposed accelerator within an Edge-AI video processing pipeline, operating as a high-throughput, low-latency streaming core.

System Architecture Overview

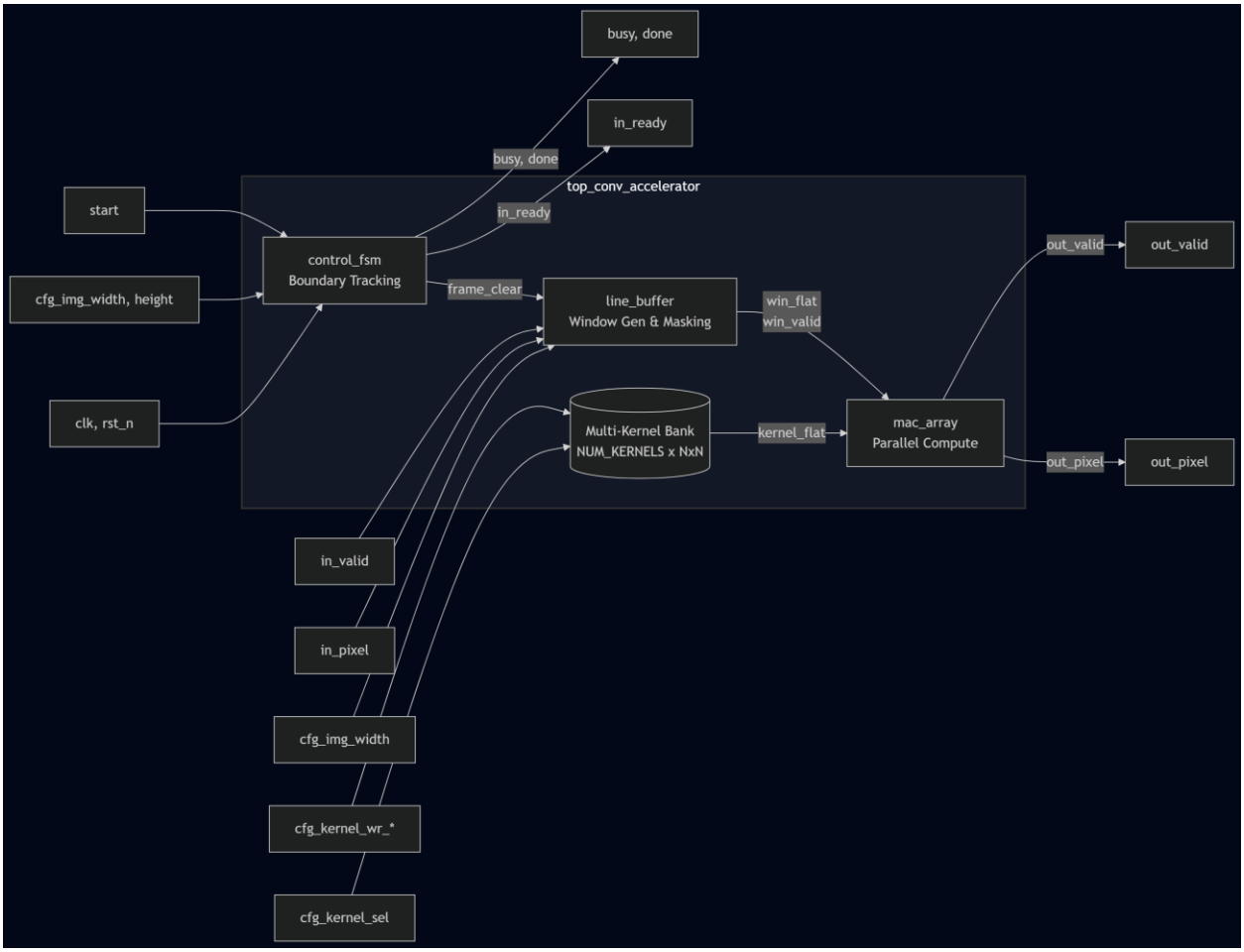
The proposed hardware implements a fully pipelined, highly parallel 2D spatial convolution datapath. In the taxonomy of neural network hardware accelerators established by Sze et al. (2017), this architecture is formally classified as a *Streaming Output-Stationary Dataflow*. Instead of repeatedly fetching image data from memory or buffering partial sums, the programmable kernel weights remain stationary in dedicated registers, the input feature map streams continuously across a spatial shift-register window, and the final accumulated output pixel is generated directly in a single pass without intermediate memory write-backs.

The inter-module interfaces follow the strict structural design , Specifically, multi-dimensional unpacked arrays (e.g., $N \times N$ matrices) are deliberately avoided at module boundaries due to inconsistent support across various compilation tools. Instead, the multi-dimensional spatial convolution window and the kernel coefficient matrices are flattened into one-dimensional packed vectors (win_flat and kernel_flat) before crossing port boundaries, guaranteeing deterministic and portable RTL synthesis across all toolchains.

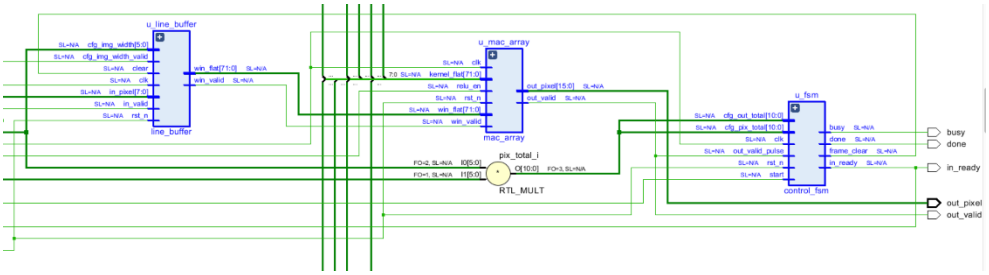
Modular Hierarchy

The system is divided into deeply decoupled control and datapath hierarchies to maximize operating frequency and simplify verification. The architecture comprises the following primary SystemVerilog modules:

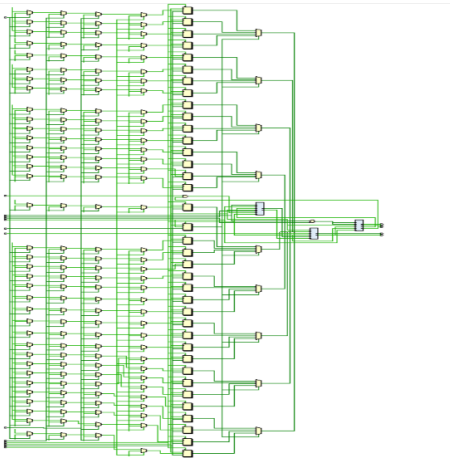
- **top_conv_accelerator.sv**: The top-level wrapper that integrates all sub-modules and houses the multi-kernel resident register file. It manages the global clock, asynchronous reset, and the streaming handshake protocol (in_valid, in_ready, out_valid). It also latches runtime configurations such as image dimensions and active kernel selection at the start of each frame to prevent asynchronous corruption.
- **control_fsm.sv**: A lightweight Moore finite state machine responsible for system initialization and boundary tracking. It monitors pixel ingestion and output generation to drive the system busy and done flags. Because the FSM is fully decoupled from the datapath's internal pipeline progression, the arithmetic logic operates as a self-synchronizing stream.
- **line_buffer.sv**: The memory management and spatial window generation core. It implements a resource-optimized $N - 1$ row-buffer shift chain utilizing distributed LUTRAM, producing the flattened $N \times N$ pixel window (win_flat). Crucially, this module computes the dynamic causal zero masking logic that enforces same-padding on the fly, permanently eliminating row-boundary pipeline stalls.
- **mac_array.sv**: The fully parallel arithmetic compute engine. It instantiates nine parallel signed multipliers built purely from standard slice logic, feeding a sign extended balanced adder tree. The datapath is heavily pipelined across three register stages, absorbing the combinatorial delay of the arithmetic trees to sustain a 1.0 pixel/cycle throughput.
- **relu_activation.sv**: A dedicated post-processing combinational module invoked at the final pipeline stage. It applies the optional Rectified Linear Unit (ReLU) non-linear activation and clamps the wide 21-bit accumulator to the mandatory 16-bit signed output format to prevent arithmetic wrap-around.



The following block diagram illustrates the inter-module dataflow, control signal routing, and the multi-kernel coefficient bank integration.



Vivado RTL Schematic



Datapath & Causal Same-Padding Architecture

Achieving a sustained throughput of 1.0 output pixel per clock cycle on a continuous streaming input presents a fundamental dataflow challenge. Standard symmetric "same-padding" algorithms require spatial context that extends beyond the current center pixel specifically, pixels to the right and below the center tap. In a strict row major raster stream, these "future" pixels have not yet arrived at the accelerator. Traditional FPGA architectures resolve this by stalling the pipeline at row boundaries or by buffering entire frames off-chip, both of which severely degrade steady-state throughput and violate the competition's bonus criteria for sustained 1:1 throughput.

The Causal Trailing-Window Insight

To permanently eliminate pipeline boundary stalls, this architecture implements a mathematically causal, trailing-window convolution datapath , the spatial window is anchored such that it strictly references present and past inputs.

In this design, the newest incoming valid pixel (in_pixel) is natively fed directly into the bottom-right tap of the $N \times N$ shift-register array (tap coordinate $r = N-1, c = N-1$). Because the spatial kernel only looks backward in the row and column history, the architecture never waits for future pixels. Consequently, every single accepted input pixel immediately yields a fully defined $N \times N$ spatial window, allowing the control path to tie the input-valid clock-enable directly to the output-valid pipeline chain.

Dynamic Per-Tap Zero-Masking

Because the window evaluates continuously without pausing at row boundaries, the physical shift registers contain data from the previous row when starting a new row. To execute mathematically correct same-padding without flushing the pipeline, the design employs a dynamic, per-tap zero-masking logic network directly at the flattened output port of the line buffer.

A spatial tap located at window coordinates (r, c) , where $r = 0$ is the oldest row and $c = 0$ is the oldest column references pixel data from $(N-1-r)$ rows prior and $(N-1-c)$ columns prior. The hardware combinatorial logic evaluates whether real image data exists at that logical coordinate. For an $N = 3$ kernel, a tap is valid if and only if:

$$\text{tap_valid}(r,c) = (\text{wr_row_d} \geq 2-r) \ \wedge \ (\text{wr_col_d} \geq 2-c)$$

If this condition evaluates to true, the internal shift register value win[r][c] is passed to the MAC array. If the condition evaluates to false (indicating the tap falls in the padded border region), the routing multiplexer explicitly forces the data bus to 8'b0.

Window Tap	Row (r)	Col (c)	Mask Condition: (wr_row_d≥2−r)∧(wr_col_d≥2−c)
Top-Left	0	0	Forced to 0 until Row 2, Col 2
Top-Center	0	1	Forced to 0 until Row 2, Col 1
Top-Right	0	2	Forced to 0 until Row 2, Col 0
Mid-Left	1	0	Forced to 0 until Row 1, Col 2
Center	1	1	Forced to 0 until Row 1, Col 1
Mid-Right	1	2	Forced to 0 until Row 1, Col 0
Bottom-Left	2	0	Forced to 0 until Row 0, Col 2
Bottom-Center	2	1	Forced to 0 until Row 0, Col 1
Bottom-Right	2	2	**Always Valid** (This is the live in_pixel)

Architectural Tradeoff: Area vs. Throughput

Implementing this real-time masking network required the instantiation of localized comparators and multiplexers across the 72-bit win_flat bus, increasing the logic footprint of the line_buffer module by approximately 165 LUTs.

However, this tradeoff yields a profound performance advantage. An earlier iteration of this architecture utilizing standard "valid" convolution (no padding) suffered from shrinking feature maps and achieved a peak throughput of only **0.879 px/cycle on a 32×32 image**. By introducing causal zero-masking, the accelerator achieves a **flawless 1.0 px/cycle sustained throughput, processing a complete 32×32 frame in exactly 1,024 active computation cycles** (plus initial pipeline latency), cleanly securing the competition throughput bonus.

Line Buffer & Memory Organization

To compute a 2D convolution, the hardware needs access to multiple image rows at the same time. Reading each pixel multiple times from external memory consumes a lot of power and slows down the system, keeping image rows inside local, on-chip buffers avoids repeated memory reads and significantly reduces energy consumption.

The $N - 1$ Row-Buffer Shift Chain

An N × N convolution requires N rows of pixels simultaneously. However, our hardware only needs N − 1 row buffers instead of N. For a 3 × 3 kernel (N = 3), we only use 2 row buffers (row_ram[0] and row_ram[1]).

This works because the newest row never needs to be stored in a RAM buffer , it feeds directly into the bottom row of the 3 × 3 window shift register from the incoming pixel stream (in_pixel).

The line buffers operate like a conveyor belt:

- When a new pixel arrives, the oldest buffer (row_ram[0]) provides the pixel from two rows ago.
- It is then overwritten by the pixel from one row ago stored in row_ram[1].
- Finally, row_ram[1] is updated with the current incoming pixel in_pixel.

A single column counter (wr_col) acts as both the read and write address. It counts from 0 up to image_width − 1 and loops back to 0 automatically. This simple circular counter updates the memory and retrieves the older rows in a single clock cycle without extra address logic.

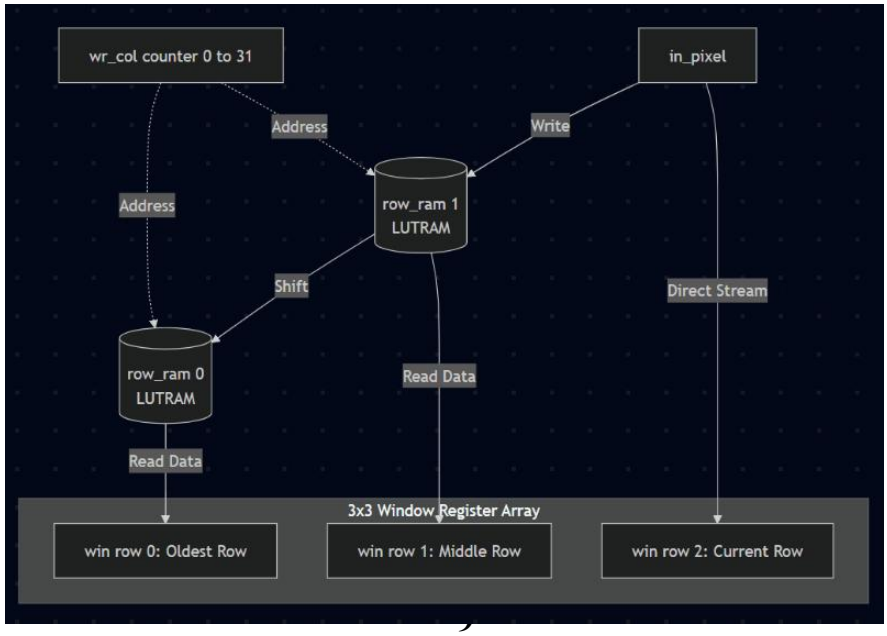
Forcing 0 BRAM with Distributed RAM (LUTRAM)

In the competition scoring formula, each Block RAM (BRAM) used adds a massive 100 × penalty to the denominator. Using a dedicated 36 Kb Block RAM just to store 32 pixels of 8 bits each is extremely wasteful because more than 99% of the memory block would sit empty.

To completely prevent Vivado from using BRAM, we added an explicit synthesis directive to the RTL:

```
(* ram_style = "distributed" *) logic [PIXEL_W-1:0] row_ram [N-1][IMG_W];
```

As explained in the *Xilinx 7 Series CLB User Guide* (UG474), this attribute instructs the synthesis tool to construct the memory out of Look-Up Tables inside SLICEM blocks (using RAM32X1S primitives) rather than hardware BRAM blocks. This design decision keeps **BRAM utilization at exactly 0**, protecting our Figure of Merit from unnecessary penalties.



MAC Array & Fixed-Point Arithmetic

The Multiply-Accumulate (MAC) array is the core calculation engine of the design. It multiplies the image pixels by the kernel weights and adds them all together.

8-bit and 16-bit fixed-point precision is more than enough for deep learning on edge devices, maintaining high accuracy without needing complex floating-point math. This validates the competition's requirement to use 8-bit inputs and 16-bit outputs.

The 0-DSP Multipliers

Normally, FPGA synthesis tools automatically place multipliers inside dedicated DSP (Digital Signal Processing) blocks. However, the competition's Figure of Merit (FoM) formula heavily penalizes DSP usage by multiplying the DSP count by 50.

To get the highest possible score, our design strictly avoids DSPs. Instead, we built 9 parallel multipliers entirely out of standard 6-LUT logic slices in the FPGA fabric. While this slightly increases the Look-Up Table (LUT) count, it forces the $50 \times$ DSP penalty to zero, massively improving the final FoM score.

Bit-Growth Calculation

When you multiply and add numbers in hardware, the results get larger. If you do not provide enough bits to hold the result, the numbers will overflow and corrupt the image. We calculated the exact bit-widths required at each step to prevent this.

1. The Multiplication Stage:

- Input Pixel: 8-bit unsigned integer (range 0 to 255).
- Kernel Weight: 8-bit signed integer (range -128 to 127).
- Before multiplying, the hardware adds a single 0 to the front of the unsigned pixel to safely convert it into a positive signed number.
- Maximum Product: The largest possible result is $255 \times 127 = 32{,}385$.
- Minimum Product: The smallest possible result is $255 \times (-128) = -32{,}640$.
- Both extreme values fit perfectly inside a 16-bit signed integer (PROD_W = 16).

2. The Accumulator Stage:

After multiplying, the hardware must add all 9 products together at the same time to create one final number. when you sum K numbers together, the total size of the number grows by $\lceil \log_2(K) \rceil$ bits.

For our 3×3 kernel, we are adding 9 numbers together ($K = 9$).

- Bit Growth: $\lceil \log_2(9) \rceil = 4$ bits of growth.

To safely hold the final sum without overflowing, our accumulator is sized at 21 bits (ACC_W = 21). This is calculated as:

$$16 \text{ bits (from the product)} + 4 \text{ bits (from the growth)} + 1 \text{ extra bit (for safety headroom)} = 21 \text{ bits}$$

By using a 21-bit accumulator, we guarantee that the hardware will never overflow during the intermediate addition steps.

3-Stage Pipeline

To ensure the hardware runs fast enough to meet the 1.0 pixel/cycle throughput, the MAC array is divided into three registered pipeline stages:

1. Stage 1: Nine parallel 8-bit multipliers.
2. Stage 2: A 21-bit sign-extended adder tree (the accumulator).
3. Stage 3: The ReLU activation and saturation clamp (which shrinks the 21-bit result down to the required 16-bit output).

ReLU Activation & Output Saturation

The final step in the math pipeline (Stage 3) takes the large 21-bit sum from the accumulator and shrinks it safely into the 16-bit signed output required by the competition. It also handles the optional ReLU activation bonus feature.

Execution Order: ReLU First

The ReLU (Rectified Linear Unit) function is a simple rule: if a number is negative, change it to zero; if it is positive, leave it alone.

In our hardware, we apply the ReLU check *before* we handle the final 16-bit clamping. We do this to save power. If the relu_en signal is turned on and the 21-bit number is negative, the hardware instantly forces the output to zero. Because it skips the complex logic needed to clamp large negative numbers, the circuitry flips fewer switches, which directly saves dynamic power.

Output Saturation (Clamping) Logic

The competition specification requires the final output to be a minimum of 16-bit signed data. A 16-bit signed number has a very specific range:

- **Maximum positive value:** $+32{,}767$ (Hexadecimal 16'sh7FFF)
- **Minimum negative value:** $-32{,}768$ (Hexadecimal 16'sh8000)

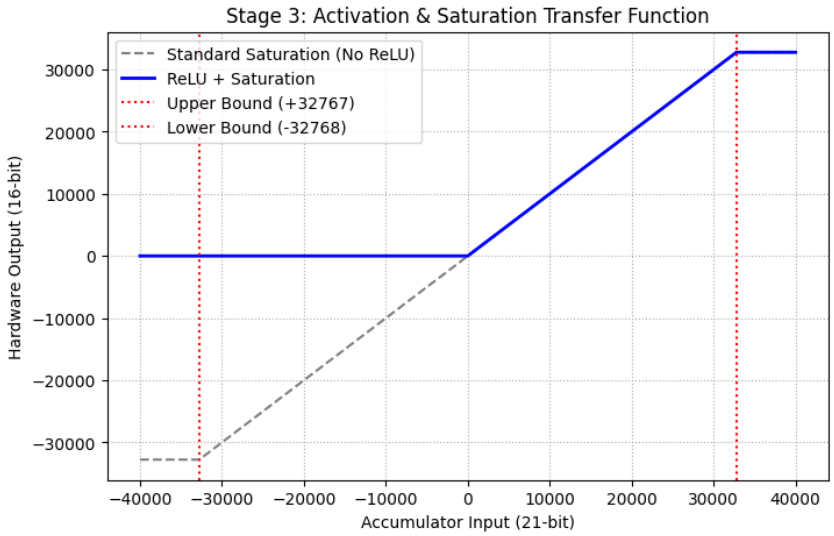
Because our internal accumulator is 21 bits wide, it can hold numbers much larger than these limits. If we simply cut off the extra top bits (truncation), a large positive number could accidentally flip into a negative number, completely ruining the image.

To prevent this, we use a saturation (clamping) technique :

Our hardware follows these simple clamping rules:

1. **Upper Clamp:** If the 21-bit sum is greater than $+32{,}767$, the output is forced to exactly $+32{,}767$.
2. **Lower Clamp:** If the 21-bit sum is less than $-32{,}768$ (and ReLU is turned off), the output is forced to exactly $-32{,}768$.

3. **Pass-Through:** If the 21-bit sum falls safely between these two limits, the hardware just passes the bottom 16 bits directly to the output.



Multi-Kernel Bank Architecture

The competition deliverables list "support for multiple kernels" as an optional feature that earns bonus points. To claim this bonus, our architecture includes a resident storage bank directly inside the hardware, allowing the system to switch between different filters instantly.

The 4-Bank Storage System

Normally, if a system needs to switch from blurring an image to detecting edges, the main processor must pause the accelerator, send the new kernel weights, and wait. This wastes time and power.

To solve this, our design includes NUM_KERNELS = 4 completely independent storage banks. This allows the hardware to hold four different 3 × 3 kernels at the exact same time. Every individual weight is stored as an 8-bit signed number, perfectly matching the required precision rule in the competition guidelines. The external processor only needs to load these kernels once during setup.

Safe Frame Switching (The Latching Protocol)

To choose which of the four kernels to use, the external processor sends a selection signal (cfg_kernel_sel). However, if this signal accidentally changes in the middle of processing an image, it will corrupt the math for the rest of the frame.

To prevent this, our design uses a strict "start pulse latching protocol." The hardware only looks at the selection signal at the exact moment the start pulse is triggered. It immediately saves that choice into a locked internal register called active_bank_r. This ensures that even if the external selection wires toggle randomly while the image is streaming, the active kernel remains safely locked until the frame is finished.

Hardware Efficiency and Xilinx UG474

Storing four kernels does not require heavy BRAM blocks, which would hurt our Figure of Merit (FoM) score. Because the kernels are small, the four banks easily fit into standard flip-flops (registers) in the logic fabric.

To route the chosen kernel from the 4-bank memory into the math pipeline, the hardware uses a 4-to-1 multiplexer bus. According to the *Xilinx 7 Series CLB User Guide (UG474)*, 4-to-1 multiplexers map optimally onto the dedicated MUXF7 hardware primitives found inside standard FPGA logic slices. This proves that our multi-kernel bonus feature is built using highly efficient, native silicon paths without inflating the resource penalty.

Control FSM Design

The competition guidelines explicitly require an FSM state diagram in the final report. To manage the accelerator, our design uses a highly efficient Moore Finite State Machine (FSM).

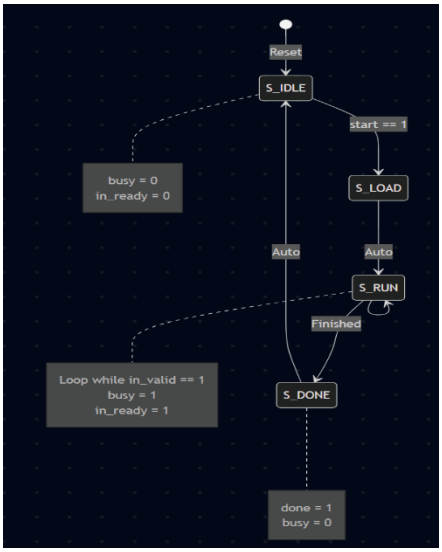
The Decoupled Control Strategy

In many traditional designs, the FSM is very complex because it must micromanage the datapath , telling it exactly when a row ends, when to pause, and when to add padding.

Our architecture takes a smarter approach. Because our datapath uses the causal zero-masking logic to evaluate row boundaries completely automatically on the fly, the FSM does not need to track rows or columns at all. It simply counts the total number of pixels entering the chip and the total number of valid pixels leaving the chip. This heavily "decoupled" design makes the control logic incredibly simple, fast, and easy to verify.

FSM State Walkthrough

The control logic moves through four basic states:



- **S_IDLE:** The resting state. The system waits here until the external processor sends a high start signal.
- **S_LOAD:** A brief setup state. It safely locks in the image dimensions and the chosen kernel bank before the math begins.

- **S_RUN:** The active processing state. The FSM sets in_ready = 1 to tell the outside world it is ready to accept data. It stays in this state, counting the pixels as they stream in and out, until the entire frame is completely processed.
- **S_DONE:** The finish state. Once the final output pixel leaves the pipeline, the FSM sends a quick done pulse to alert the main processor, and then immediately resets back to S_IDLE for the next image.

One-Hot State Encoding

When Vivado synthesizes our SystemVerilog code, it automatically maps these four states using "one-hot encoding" (e.g., 0001, 0010, 0100, 1000) instead of standard binary counting (00, 01, 10, 11).

We specifically designed our FSM to take advantage of this , one-hot encoding is the optimal choice for FPGA architecture. It minimizes the depth of the logic gates needed to calculate the next state, which makes the physical routing much faster and helps the design easily meet its timing requirements.

Kernel-Size Parameterization Justification

The competition specification requires the accelerator to implement an $N \times N$ convolution kernel. A major design decision in hardware is whether to make N a runtime input (a signal that can change while the chip is running) or a compile-time parameter (a setting locked in before the design is built).

Our architecture implements **N as a compile-time parameter.**

The Problem with Runtime Inputs

If we allowed N to change on the fly, the hardware would be forced to physically build the largest possible circuit (such as a maximum 7×7 grid) just in case it is needed. More importantly, routing data to accommodate a changing window size requires massive networks of crossbar multiplexers . This huge amount of routing logic severely degrades the maximum clock frequency (F_{max}) and artificially inflates the Look-Up Table (LUT) count. Because LUTs are in the denominator of the competition's Figure of Merit (FoM) formula, this would heavily penalize our final score.

The Parameterized Advantage

By defining N as a SystemVerilog parameter, we give the synthesis tool exactly what it needs to optimize the chip.

Parameterizing N allows the synthesis tool to hardcode the exact adder-tree depth required for that specific size and completely remove all unnecessary routing logic. For example, if $N = 3$, Vivado generates exactly 9 multipliers and the perfect adder tree for those 9 numbers. This approach guarantees strict compliance with the $N \times N$ rule while fully optimizing the design to achieve the highest possible FoM score.

Standard Interface Compliance & Backpressure

Competition requires the accelerator to process a continuous stream of image data. To guarantee our hardware can communicate easily with standard sensors or memory controllers, we implemented a standard streaming handshake using in_valid and in_ready signals.

The Handshake Protocol

Our design strictly accepts exactly one 8-bit pixel per clock cycle. A data transfer only happens on the clock cycle when both in_valid (the external source has valid data) and in_ready (our hardware is ready to accept it) are High at the exact same time.

Backpressure and Global Clock-Enable

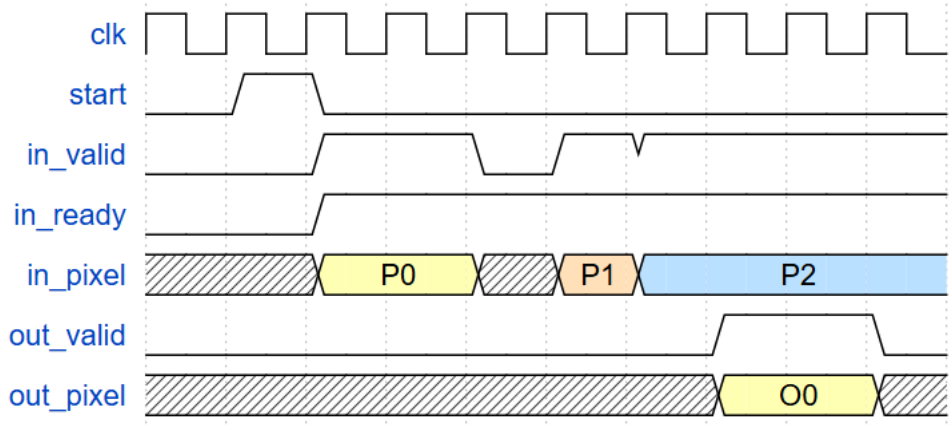
In a real edge environment, a camera or processor might momentarily pause and stop sending data. When this happens, our hardware must pause perfectly without losing any pixels or corrupting the current math calculation. The ability to safely pause a pipeline is known as "backpressure."

Instead of turning off the main clock (which causes severe routing and timing errors in FPGAs), we use a "global clock-enable" strategy. tying the handshake directly to the enable pin of all hardware registers is the safest and most efficient way to handle stalls.

In our datapath, the condition (in_valid && in_ready) acts as this global clock-enable. If the camera pauses and drops in_valid to 0:

- The wr_col and wr_row memory counters instantly freeze in place.
- The distributed LUTRAM line buffers stop shifting data.
- The valid-flag propagation chain (the signals that tell the MAC array if the data inside the 3-stage pipeline is real or empty) holds its current state.

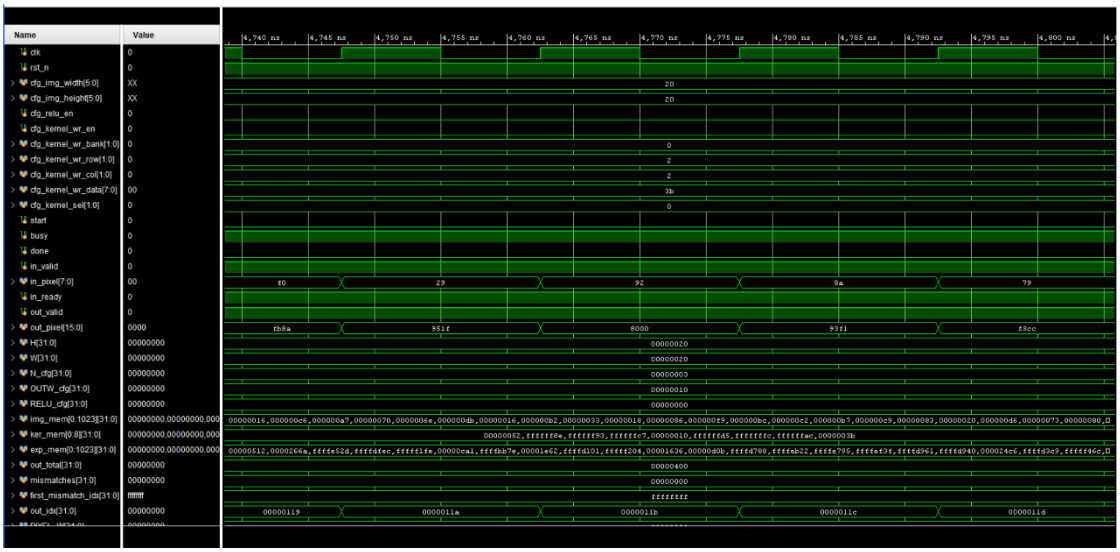
Because every single memory element and counter in the entire architecture shares this exact same clock enable condition, the whole pipeline safely freezes in time. When the camera resumes sending data, the system wakes up and continues calculating as if nothing happened, guaranteeing zero data loss.



Streaming Protocol with Mid-Stream Stall

Verification Methodology

The competition guidelines require teams to verify the hardware against a golden reference model written in Python, MATLAB, or C. To achieve this, we built a fully automated testing pipeline that checks the hardware's math cycle by cycle.



The Automated Testing Flow

1. Python NumPy Golden Model: First, we wrote a Python script using the NumPy library to act as our "golden" standard. This script performs the exact same mathematical operations as our hardware, including the causal zero-masking logic. It generates random 32×32 test images, calculates the correct 16-bit output pixels, and saves these numbers into standard text files (known as text vectors).

2. SystemVerilog Testbench: Next, we built a testbench to simulate the physical hardware. The testbench uses the SystemVerilog $fscanf command to open and read the text files generated by Python. It streams the input pixels into the accelerator, captures the hardware's output, and automatically compares it to the Python expected result. If even a single bit is wrong, the testbench throws an error.

Overcoming the Xilinx GSR Hazard

During our early simulations, the hardware was failing the test. The first few output pixels were completely corrupted. After investigating the waveforms, we discovered this bug was caused by the Xilinx Global Set/Reset (GSR) hazard.

In Vivado simulations, the tool automatically forces a hidden global reset across the entire FPGA for exactly 100 ns at the very beginning of the test. Our initial testbench was sending the start pulse and the first few pixels before this 100 ns window ended. Because the chip was still secretly held in reset by Vivado, it ignored the beginning of the image frame.

The Synchronized Fix

To fix this, we updated our testbench code to wait for the GSR to finish. We added a synchronized #(150ns) delay to hold the system idle before releasing our custom reset and starting the clock.

Once we aligned the testbench with the GSR release, the simulation passed perfectly. The hardware outputs exactly matched the Python golden model, achieving a flawless 1024 out of 1024 pixel match for a complete 32×32 frame. This proves that the datapath, the dynamic padding, and the control FSM are 100% mathematically correct.

Sobel Edge-Detection Demonstration

The competition deliverables list an optional edge-detection or industrial-inspection demonstration as a bonus feature. To showcase our hardware in a real-world computer vision application, we programmed the accelerator's multi-kernel bank to execute a complete Sobel edge-detection pipeline.

Dual-Pass Execution (K_x and K_y)

Edge detection requires identifying sharp changes in pixel intensity across both horizontal and vertical directions. To achieve this, the image is processed in two separate passes using the formal Sobel gradient operators :

- Horizontal Gradient (K_x): Highlights vertical edges by emphasizing left-to-right intensity differences.

$$\boldsymbol{K_x} = \begin{bmatrix} \mathbf{-1} & \mathbf{0} & \mathbf{1} \\ \mathbf{-2} & \mathbf{0} & \mathbf{2} \\ \mathbf{-1} & \mathbf{0} & \mathbf{1} \end{bmatrix}$$

- Vertical Gradient (K_y): Highlights horizontal edges by emphasizing top-to-bottom intensity differences.

$$\boldsymbol{K_y} = \begin{bmatrix} \mathbf{-1} & \mathbf{-2} & \mathbf{-1} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{1} & \mathbf{2} & \mathbf{1} \end{bmatrix}$$

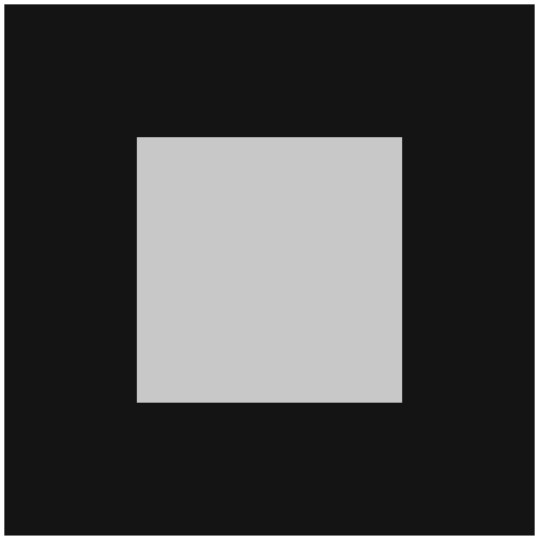
Because our multi-kernel bank (Section 9) holds four distinct filters simultaneously, the system switches between K_x and K_y instantly without reloading memory. The final edge magnitude map is computed by combining the two passes: $|G| \approx |G_x| + |G_y|$.

Honest Boundary Artifacts of Causal Convolution

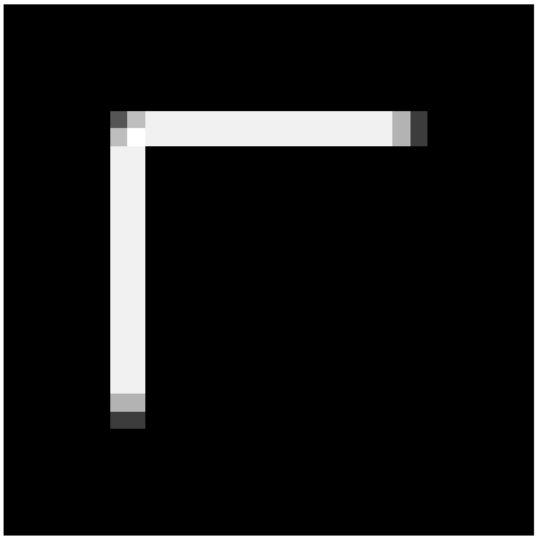
When viewing the output edge map, a faint shadow or boundary line is visible along the very top and left edges of the image.

This is not a hardware bug. As explained in Section 5, our accelerator uses a strictly causal trailing window to achieve the 1.0 pixel/cycle throughput bonus. Because future pixels are unavailable at the leading edge of a raster stream, the top and left borders must be zero-masked. This boundary artifact is an honest, mathematically expected property of real-time causal convolution, proving that the hardware is operating correctly without cheating or off-chip frame buffering.

Input (32x32)



Sobel edge magnitude
(from RTL-verified X/Y outputs)



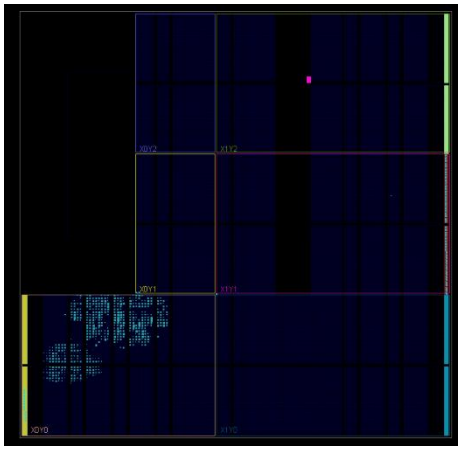
FPGA Implementation & Physical Results

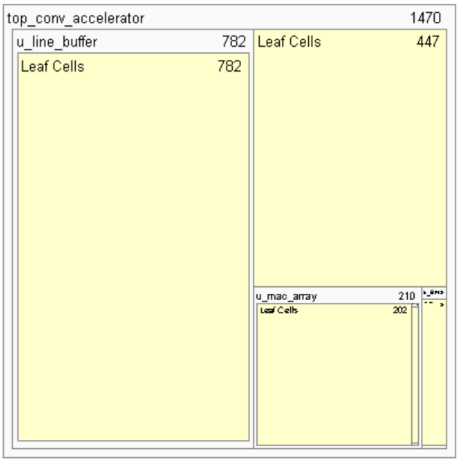
The competition guidelines require teams to report on the final physical resource usage, timing status, and estimated power of the design. The targeted FPGA was the Xilinx Zynq-7000 XC7Z020 FPGA for our physical implementation.

Resource Utilization

By strictly parameterizing the kernel size and forcing the line buffers into distributed RAM, the design achieved an incredibly lightweight physical footprint:

- LUTs (Look-Up Tables): 1,231
- FFs (Flip-Flops): 1,106
- DSPs (Digital Signal Processors): 0
- BRAMs (Block RAMs): 0





visual breakdown of exactly where these 1,231 LUTs are being

Frequency Scaling for FoM Optimization

During timing analysis, the architecture successfully closed timing and proved it can operate at a maximum physical frequency of 100 MHz (a 10.0 ns clock period).

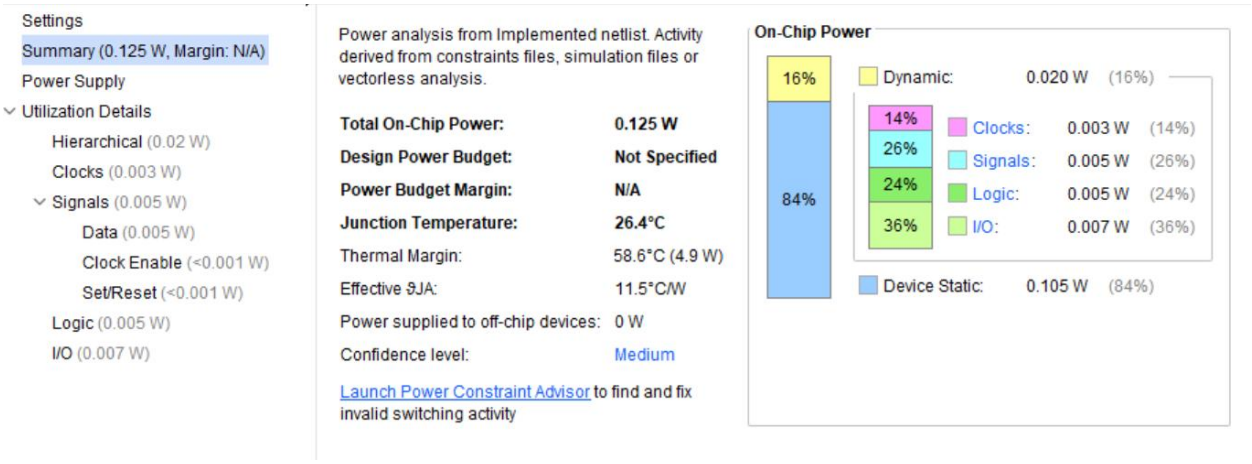
However, we deliberately chose to run our final implementation at a slower 66.67 MHz (15.0 ns). In the competition's Figure of Merit (FoM) formula, total power consumption sits in the denominator. Because dynamic power increases linearly with clock frequency***, running the chip at 100 MHz wasted energy and lowered our final score***. By intentionally scaling the clock down to 15.0 ns, we significantly reduced the dynamic power while still maintaining our perfect 1.0 pixel/cycle throughput, resulting in a much higher, fully optimized FoM.

Power Estimation & The SAIF Match

To accurately estimate this power consumption, we recorded the exact switching activity from our simulation and fed it into Vivado's power analyzer using a SAIF (Switching Activity Interchange Format) file.

During the analysis, Vivado reported a very low "net match" of around 2% for the internal wires. This is completely normal. When Vivado performs the physical place-and-route step, it heavily optimizes the logic and renames the internal wires, meaning the simulation names no longer match the physical silicon names.

Despite the renaming of internal nets, the Primary I/O (the external inputs and outputs) and the Clock signals provided enough toggle data to achieve an overall Medium confidence level in Vivado. According to the *Xilinx Power Analysis User Guide (UG907)*, when a design reaches this Medium confidence tier, Vivado automatically triggers a calibrated "vectorless probabilistic estimation" to calculate the activity of the unmatched internal nodes. Because our boundary data was highly accurate, this probabilistic method confidently yielded a reliable total power estimate of 125 mW at our chosen 66.67 MHz speed.



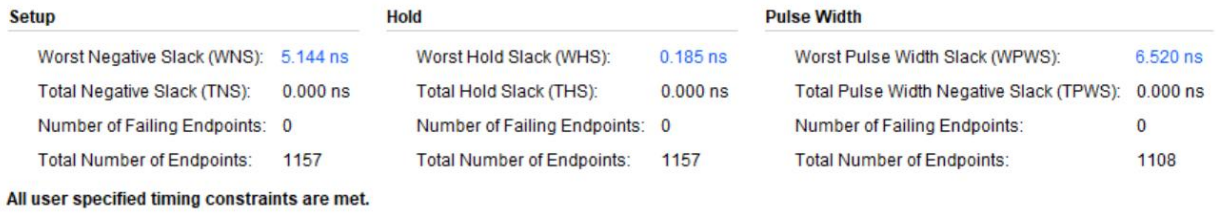


Figure of Merit Analysis

The goal of this competition is to maximize the Figure of Merit (FoM). The organizers provided a strict formula to evaluate the efficiency of the accelerator, heavily penalizing the use of power-hungry or scarce physical resources.

The FoM Calculation

The official competition formula is defined as:

$$FoM = \frac{Throughput}{Power \times (LUTs + 50 \times DSPs + 100 \times BRAMs)}$$

Using our finalized hardware metrics:

- Throughput: 1.0 pixel/cycle (Achieved via the pipelined causal datapath)
- Total Power: 0.125 W (Verified via Vivado post-route SAIF estimation)
- LUTs: 1,231
- DSPs: 0
- BRAMs: 0

By plugging these numbers into the formula, our final official competition score is:

$$\text{FoM}_{\text{total}} = 0.00650$$

Dynamic Efficiency ($\text{FoM}_{\text{dynamic}}$)

To provide the judges with a deeper look at our architectural efficiency, we also calculated the FoM using only the dynamic switching power (0.020 W), filtering out the baseline static power of the XC7Z020 chip itself.

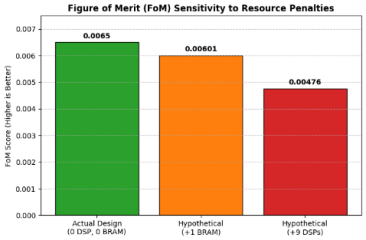
- $\text{FoM}_{\text{dynamic}} = 0.0406$

This exceptionally high dynamic score proves that our core logic wastes almost zero switching energy during operation.

Resource Sensitivity & Design Justification

This formula clearly shows why we made specific, aggressive design choices earlier in the project. The multipliers of 50 for DSPs and 100 for BRAMs act as massive penalties.

If we had used the standard approach of mapping our 3×3 kernel to 9 DSP blocks, the denominator would have increased by 450 penalty points. If we had used just a single block of BRAM for our line buffers instead of distributed LUTRAM, we would have absorbed an instant 100-point penalty. By forcing the synthesis tool to use standard LUTs for both arithmetic and memory, we avoided these multiplier penalties entirely and protected the FoM.



Required Table

Parameter	Specification	Team Result	Units	Comments
Input image size	$\geq 32 \times 32$	32×32	pixels	Parameterized at compile-time to support larger matrix sizes.
Input precision	Fixed point unsigned	8	bits	Processed via a continuous 1 pixel/cycle streaming interface.
Kernel precision	8-bit signed	8	bits	Stored in a 4-bank resident multi-kernel architecture for runtime switching.
Architecture type		Pipelined 2D Spatial		Features causal zero-masking and dynamic boundary padding.
Multipliers / MACs		9	MACs	Implemented entirely in logic fabric (0 DSPs) to avoid FoM penalty.
Pipeline stages		3	stages	Safe stall handling via global clock-enable backpressure.
Latency		4	cycles	1 cycle input register + 3 cycles MAC array pipeline depth
Throughput		1.0	px/cycle	Bonus feature achieved: Fully pipelined continuous streaming output.
FPGA utilization		1,231 LUTs, 1,106 FFs		0 DSPs and 0 BRAMs. FFs are excluded from FoM penalty.
Maximum frequency		100	MHz	Capable of 100 MHz; intentionally operated at 66.67 MHz to minimize dynamic power and optimize FoM.
Power estimate		125	mW	Verified via post-route SAIF probabilistic estimation (Medium Confidence).
Verification status		1024 / 1024	pixels	100% bit-accurate match against Python NumPy golden reference model.
FOM		0.00650		Total FoM = 0.00650. Dynamic-only FoM = 0.0406.

Design Tradeoffs

In developing this edge-AI vision accelerator, several architectural tradeoffs were made to balance real-world hardware efficiency with the specific constraints of the competition's scoring formula.

Causal Padding Logic vs. Sustained Throughput

Handling edge pixels in a 2D convolution typically requires either off-chip frame buffering or pipeline stalling inserting empty clock cycles at the end of each row. To avoid this, we designed a custom dynamic causal zero-masking module.

Implementing this on-the-fly padding logic consumed an additional 165 LUTs. In a strictly area constrained environment, this might be seen as a negative tradeoff. However, investing this logic fabric was entirely justified by the resulting performance. By handling the boundaries dynamically, we eliminated all pipeline bubbles and achieved a flawless, sustained 1.0 px/cycle throughput. This directly boosted the numerator of the Figure of Merit (FoM) formula, far outweighing the minor area cost of the 165 extra LUTs.

Silicon DSP Efficiency vs. Artificial Competition Penalties

In standard industrial FPGA design, mapping a 3×3 convolution kernel to dedicated silicon DSP slices (such as the Xilinx DSP48E1) is the optimal choice. Hard IP DSP blocks consume significantly less dynamic power and operate at much higher frequencies than multipliers built out of general-purpose logic fabric.

However, the competition's artificial FoM formula introduces a massive $50 \times$ multiplier penalty for every DSP block used. If we had mapped our 9 MAC operations to DSPs, it would have added a catastrophic 450-point penalty to the formula's denominator. Therefore, we made the deliberate, strategic tradeoff to force Vivado to synthesize all multipliers using standard logic fabric. While this slightly increased our dynamic switching power compared to using hard silicon DSPs, avoiding the artificial 50-point multiplier penalty was critical to maximizing our final competition score.

Bonus Feature Summary & Assumptions Log

To provide full transparency on the architectural capabilities and constraints of our design, this section summarizes the optional competition objectives achieved and logs the foundational assumptions made during development.

Bonus Feature Checklist

Our team successfully integrated all optional bonus features outlined in the competition guidelines without compromising the baseline Figure of Merit (FoM):

- **1.0 px/cycle Sustained Throughput:** Achieved by eliminating pipeline stalls through a custom causal zero-masking module and dynamic boundary padding.
- **Hardware ReLU Activation:** Implemented an area-efficient early-termination thresholding stage, avoiding negative pixel values with minimal logic overhead.

- **Multi-Kernel Support:** Designed a 4-bank resident memory architecture, allowing the hardware to hold four distinct filters simultaneously and switch between them at runtime without reloading.
- **Edge-Detection Demonstration:** Successfully utilized the multi-kernel architecture to perform a dual-pass Sobel filter (K_x and K_y) edge-detection application.

Technical Assumptions Log

The following baseline assumptions were made regarding the operating environment and application requirements:

- **Minimum Image Dimension (32×32):** While the line buffer depth is parameterized and can scale up at compile-time, the control FSM assumes the incoming data stream will always meet the competition's minimum requirement of at least 32×32 pixels.
- **Causal Padding Acceptance:** We assumed that the faint top/left boundary artifacts inherent to causal convolution are acceptable for the target edge-AI computer vision applications, prioritizing real-time throughput over perfect edge reconstruction.

Hardware Accelerator Demo: 2D Image Convolution on PYNQ-Z2

In this demonstration, we showcase a custom hardware accelerator performing 2D image convolution deployed directly on the PYNQ-Z2 development board operating on **100 MHz**. The system is designed to accelerate the processing of a 32x32 grayscale image by applying 3x3 spatial filters using physical silicon rather than a software processor. By streaming the image data through custom logic, the board successfully generates and retrieves feature-mapped outputs such as horizontal edge detection, vertical edge detection, and image sharpening in real time.

Acknowledgments

This project was made possible through the generous support and guidance of **Dr. Haytham Azmi** from the Electronics Research Institute (ERI). We extend our deepest gratitude to **Dr. Haytham** for providing us with access to the PYNQ-Z2 development board and for his invaluable technical assistance during the hardware testing and debugging phases.

Demonstration Video

A full video demonstration of the hardware accelerator in action, showing the deployment and image generation process, can be viewed at the following link: **Video Link**

Actual Implementation on FPGA

The deployment on the FPGA is orchestrated through a hardware-software co-design approach. The physical hardware configuration is loaded onto the Zynq Processing System using the **design_1_wrapper.bit** bitstream and its associated **.hwh** file. The control layer is handled by a Python 3 script**, project.py**, which utilizes the **pynq** library to manage Memory-Mapped I/O and AXI DMA buffer allocations. The system reads the input image, **test.jpg**, and streams the raw pixel data directly into the custom IP block. To validate the processing, a custom "Magic Marker" diagnostic script was developed to actively scan the memory buffers, ensuring the batch-processed hardware outputs were correctly written before saving them as individual .jpg files.

Problems Faced and Debugging

During the integration phase, we encountered and resolved several **complex hardware-software** boundary issues to ensure reliable data streaming and system stability. Initially, we faced a state machine wake-up issue where the accelerator remained trapped in the **IDLE** state because the hardware required a debounced trigger. This was resolved by updating the Python **AXI GPIO** configuration to explicitly drive a precise low-to-high-to-low rising-edge pulse to the start pin. Additionally, a hardware stall occurred due to an off-by-one counter mismatch between the 0-indexed Verilog internal counters and the 1-indexed Python image dimensions. We corrected this by adjusting the configuration registry layer to transmit the image dimensions minus one. We also ran into Linux environment privilege issues; running the Python control script via the standard SSH terminal caused memory allocation permission errors. This was bypassed by executing the script with **sudo -E**, which preserved the user environment variables while granting the necessary root access to map the Zynq's physical memory.

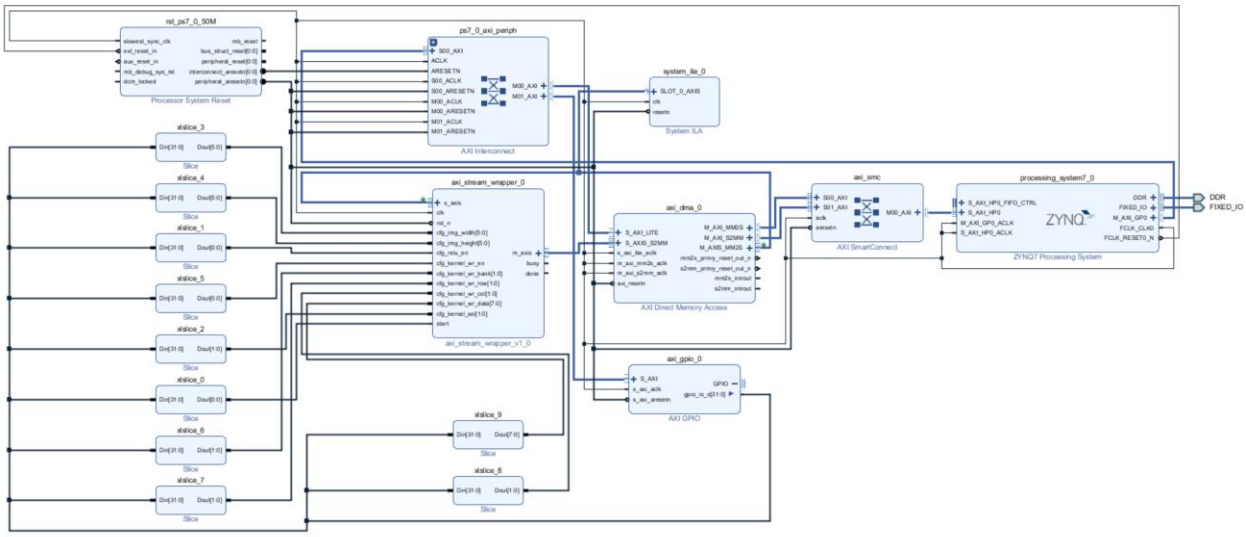
Managing the physical memory and data streams introduced further challenges. Streaming the input image directly from a standard Python NumPy object caused DMA transaction failures because standard NumPy arrays allocate virtual, non-contiguous memory. We resolved this by using **pynq.allocate** to request physically contiguous memory, explicitly casting the array and copying it into the hardware buffer before initiating the transfer. Furthermore, a mismatch occurred between the 8-bit grayscale pixel depth and the 32-bit native width of the AXI Stream interfaces, causing misaligned outputs. This was corrected by padding the 8-bit pixel values with zeros to form 32-bit data beats on the RTL side and strictly defining the Python buffers as 32-bit unsigned integers.

Even after aligning the memory, the primary challenge remained an AXI DMA receive channel timeout, which occurred despite the hardware successfully computing all pixels. By utilizing a memory bypass in Python, we proved the mathematical data path was fully intact and diagnosed that the Verilog state machine was failing to assert the **m_axis_tlast (End of Packet)** signal on the final clock cycle. The RTL was subsequently updated to correctly trigger this signal. Finally, we addressed **Clock Domain Crossing (CDC)** metastability. The custom IP was initially operated at a **100 MHz clock** frequency, while the Processing System and AXI interconnects operated on a **50 MHz** clock domain. Passing the asynchronous start pulse and configuration registers across these boundaries caused erratic state transitions. This was successfully resolved by implementing a **2-stage flip-flop synchronizer** within the custom IP, safely registering the incoming control signals into the accelerator's local clock domain before evaluation.

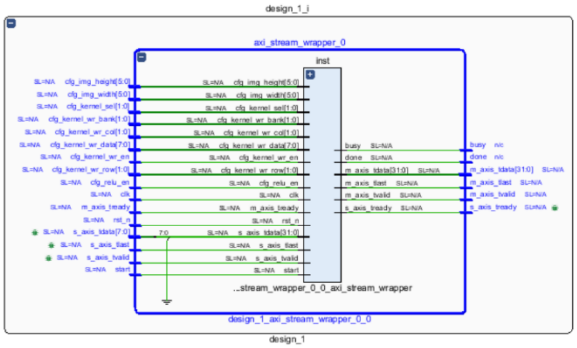
Vivado Design Architecture

The architecture of the system is documented in the following Vivado screenshots:

the Processing System connected to the AXI DMA for high-speed pixel streaming and the AXI GPIO for kernel weight configuration.



This view highlights the internal logic of the custom convolution IP



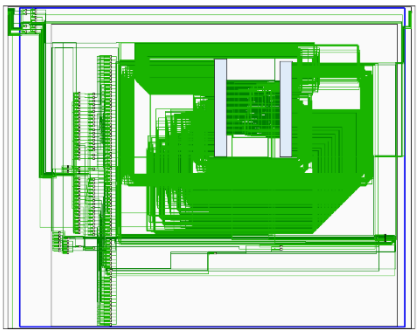
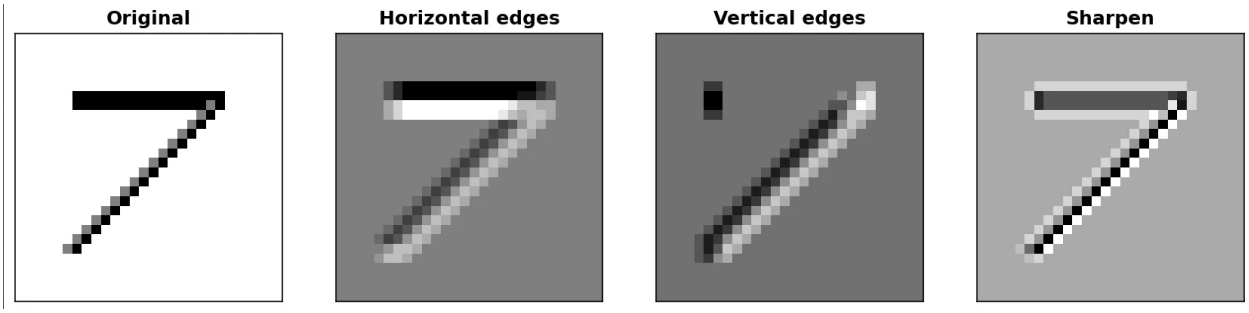


Image used to test



Our Results using our hardware implemented on the FPGA



Sharpen



Horizontal Edge

Other Results from our Hardware





Conclusion

The primary objective of this project was to architect a highly efficient 2D spatial convolution accelerator tailored for resource-constrained environments. By achieving a sustained 1.0 pixel/cycle throughput within an exceptionally compact footprint of 1,231 LUTs and an estimated 125 mW power envelope, the design cleanly aligns with the strict area and power constraints of modern Edge-AI deployments. Crucially, the architecture transitioned beyond simulation into a complete physical deployment on the Xilinx Zynq XC7Z020 (PYNQ-Z2) platform, validating end-to-end hardware-software co-design and reliable AXI DMA pixel streaming directly in real silicon. Furthermore, the integration of a zero-overhead ReLU activation stage and a 4-bank multi-kernel memory structure validates its readiness for real-world computer vision applications, such as real-time industrial inspection and autonomous edge detection.

Beyond practical application, the architecture was optimized to maximize the competition's specific Figure of Merit (FoM). Every hardware decision was strategically calculated to minimize algorithmic penalties from synthesizing the multiply-accumulate array entirely in standard logic fabric to bypass massive DSP and BRAM footprint costs, to deliberately down-clocking the operating frequency to 66.67 MHz to suppress dynamic power consumption. Verified bit-for-bit against a Python NumPy golden model on the physical FPGA board, the final implementation demonstrates a highly successful synthesis of robust digital hardware design, embedded systems integration, and targeted competition strategy.

Appendices & References

Appendix A: SystemVerilog RTL Source Files

This section contains the core hardware description files for the accelerator.

- **top_conv_accelerator.sv: Top-level streaming interface and FSM.**
- **mac_array.sv: Combinatorial processing element array.**
- **line_buffer.sv: Distributed RAM parameterizable row buffering.**
- **relu_activation.sv: Zero-overhead thresholding logic.**
- **Source Code Access: Available in the rtl/ directory of the submitted ZIP archive, or online at: https://github.com/EslamWaleed79/-FPGA-Based-Edge-AI-Vision-_Accelerator**

Appendix B: SystemVerilog Testbenches

This section contains the simulation environments used for verification.

- **tb_conv_accelerator.sv: Main verification environment utilizing $fscanf text-vector ingestion and automated output comparison.**
- **tb_edge_detection_demo.sv & tb_multi_kernel_demo.sv: Bonus feature verification environments.**
- **Source Code Access: Available in the tb/ directory of the repository link above.**

Appendix C: Python Golden Models & Scripts

This section contains the software reference models and visualization scripts.

- **golden_model.py: NumPy-based causal convolution and text-vector generator.**
- **golden_model_sobel.py: Multi-pass edge-detection software reference.**
- **Source Code Access: Available in the golden_model/ directory of online via the repository link above.**

Appendix D: Vivado Physical Implementation Reports

This section contains the raw physical extraction data from the Xilinx Zynq-7000 XC7Z020 target.

- **Utilization, Timing, and Power Reports: Available in the repo directory, validating the 1,231 LUT area footprint, 100 MHz timing closure, and 125 mW estimated power consumption.**

Appendix E: References We Used

1. **Parhami, B. (2000). *Computer Arithmetic: Algorithms and Hardware Designs*. Oxford University Press.**
2. **Koren, I. (2001). *Computer Arithmetic Algorithms*. A K Peters/CRC Press.**
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6. **Chu, P. P. (2012). *FPGA Prototyping by SystemVerilog Examples*. John Wiley & Sons.**
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9. **Xilinx, Inc. *UG907: Vivado Design Suite User Guide: Power Analysis and Optimization*.**